Case Study 5: ATM Transaction Sequence Simulation

1. Problem Statement

To develop a Python program that simulates ATM transactions by processing deposits and withdrawals. The program maintains the account balance, validates transactions, enforces withdrawal limits, rejects invalid operations, and displays a summary of all transactions.

2. Objectives

- To simulate basic ATM operations.
- To process deposits and withdrawals.
- To maintain the account balance after every transaction.
- To reject invalid transactions and withdrawals exceeding the available balance or withdrawal limit.
- To generate a transaction summary.

3. Assumptions

The following values are assumed for this assignment.

Component	Value
Initial account balance	₹5,000
Maximum withdrawal per transaction	₹2,000
Minimum deposit amount	₹1
Minimum withdrawal amount	₹1

Allowed transactions	Deposit and withdrawal

These are sample rules for the assignment, not actual bank policies.

4. Algorithm

1. Start the program.
2. Initialize the account balance to ₹5,000.
3. Ask the user to enter the number of transactions.
4. For each transaction, enter the transaction type and amount.
5. If the transaction is a deposit, add the amount to the balance.
6. If it is a withdrawal, check the available balance and withdrawal limit.
7. Reject invalid transactions.
8. Record each transaction and its status.
9. Display the updated balance after each transaction.
10. Display the final balance and transaction summary.
11. Stop the program.

5. Flowchart

Start
Initialize balance = ₹5,000
Enter transaction type and amount
Is it a deposit?
Deposit

Add amount to balance

Withdrawal

Validate limit and balance

Update balance and record transaction status
Display transaction summary and final balance
Stop

6. Pseudocode

```
None
START
```

```
SET balance = 5000
SET transaction_history = empty list

INPUT number_of_transactions

FOR each transaction:
    INPUT transaction_type
    INPUT amount

    IF amount <= 0:
        DISPLAY "Invalid amount"

    ELSE IF transaction_type is "deposit":
        ADD amount to balance
        RECORD successful deposit

    ELSE IF transaction_type is "withdraw":
        IF amount > 2000:
            DISPLAY "Withdrawal limit exceeded"
        ELSE IF amount > balance:
            DISPLAY "Insufficient balance"
        ELSE:
            SUBTRACT amount from balance
            RECORD successful withdrawal

    ELSE:
        DISPLAY "Invalid transaction type"

    DISPLAY current balance

DISPLAY transaction summary
DISPLAY final balance

STOP
```

7. Complete Python source code

```
# ATM Transaction Sequence Simulation


balance = 5000.0

withdrawal_limit = 2000.0

transaction_history = []


print("===== WELCOME TO PYTHON ATM =====")

print("Initial Balance: Rs.", balance)


number_of_transactions = int(

    input("Enter number of transactions: ")

)


if number_of_transactions < 0:

    print("Invalid number of transactions.")


else:

    for i in range(number_of_transactions):

        print("\nTransaction", i + 1)


        transaction_type = input(

            "Enter deposit or withdraw: "
```

```
).strip().lower()

amount = float(input("Enter amount: Rs. "))

if amount <= 0:
    status = "Rejected: Invalid amount"

elif transaction_type == "deposit":
    balance += amount
    status = "Successful deposit"

elif transaction_type == "withdraw":
    if amount > withdrawal_limit:
        status = "Rejected: Withdrawal limit exceeded"
    elif amount > balance:
        status = "Rejected: Insufficient balance"
    else:
        balance -= amount
        status = "Successful withdrawal"

else:
    status = "Rejected: Invalid transaction type"

transaction_history.append(
```

```
        (transaction_type, amount, status, balance)
    )

    print("Status:", status)
    print("Current Balance: Rs.", round(balance, 2))

# Display transaction summary
print("\n===== TRANSACTION SUMMARY =====")

for transaction in transaction_history:
    kind, amount, status, current_balance = transaction

    print(
        kind.title(),
        "| Amount: Rs.", round(amount, 2),
        "|", status,
        "| Balance: Rs.", round(current_balance, 2)
    )

successful_deposits = sum(
    1 for t in transaction_history
    if t[0] == "deposit" and t[2] == "Successful deposit"
)
```

```
    successful_withdrawals = sum(
        1 for t in transaction_history
        if t[0] == "withdraw" and t[2] == "Successful withdrawal"
    )

    rejected_transactions = sum(
        1 for t in transaction_history
        if t[2].startswith("Rejected")
    )

    print("\nSuccessful Deposits:", successful_deposits)
    print("Successful Withdrawals:", successful_withdrawals)
    print("Rejected Transactions:", rejected_transactions)
    print("Final Balance: Rs.", round(balance, 2))
    print("==================================")
```

```
None
```

8. Test cases and expected output

Use the following inputs to test your program.

Transaction	Type	Amount	Expected result	Balance

1	Deposit	₹1,000	Successful	₹6,000
2	Withdraw	₹1,500	Successful	₹4,500
3	Withdraw	₹3,000	Rejected: limit exceeded	₹4,500
4	Withdraw	₹5,000	Rejected: limit exceeded	₹4,500
5	Deposit	₹500	Successful	₹5,000

Sample input:

```
None
Enter number of transactions: 3

Transaction 1
Enter deposit or withdraw: deposit
Enter amount: Rs. 1000

Transaction 2
Enter deposit or withdraw: withdraw
Enter amount: Rs. 1500

Transaction 3
Enter deposit or withdraw: withdraw
Enter amount: Rs. 3000
```

Expected final summary:

None

```
===== TRANSACTION SUMMARY =====
Deposit | Amount: Rs. 1000.0 | Successful deposit | Balance: Rs.
6000.0
Withdraw | Amount: Rs. 1500.0 | Successful withdrawal | Balance:
Rs. 4500.0
Withdraw | Amount: Rs. 3000.0 | Rejected: Withdrawal limit exceeded
| Balance: Rs. 4500.0

Successful Deposits: 1
Successful Withdrawals: 1
Rejected Transactions: 1
Final Balance: Rs. 4500.0
================================
```

9. Output screen shot

```python
balance = 5000.0
withdrawal_limit = 2000.0
transaction_history = []

print("===== WELCOME TO PYTHON ATM =====")
print("Initial Balance: Rs.", balance)

number_of_transactions = int(
    input("Enter number of transactions: ")


if number_of_transactions < 0:
    print("Invalid number of transactions.")

else:
    for i in range(number_of_transactions):
        print("\nTransaction", i + 1)

        transaction_type = input(
            "Enter deposit or withdraw: "
        ).strip().lower()

        amount = float(input("Enter amount: Rs. "))

        if amount <= 0:
            status = "Rejected: Invalid amount"

        elif transaction_type == "deposit":
            balance += amount
            status = "Successful deposit"

        elif transaction_type == "withdraw":
            if amount > withdrawal_limit:
                status = "Rejected: Withdrawal limit exceeded"
            elif amount > balance:
                status = "Rejected: Insufficient balance"
            else:
                balance -= amount
                status = "Successful withdrawal"

        else:
            status = "Rejected: Invalid transaction type"

        transaction_history.append(
            (transaction_type, amount, status, balance)
        )

        print("Status:", status)
        print("Current Balance: Rs.", round(balance, 2))

    # Display transaction summary
    print("\n===== TRANSACTION SUMMARY =====")
```

25°C
Partly cloudy
Search

```python
for transaction in transaction_history:
    kind, amount, status, current_balance = transaction

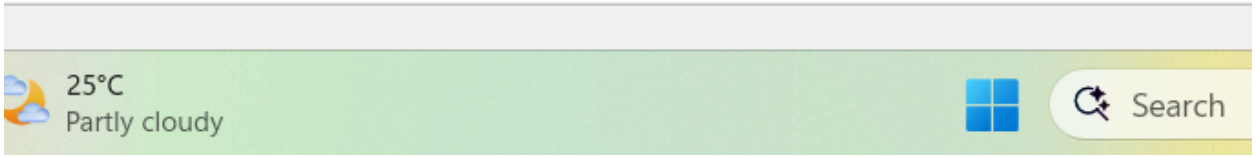
    print(
        kind.title(),
        "| Amount: Rs.", round(amount, 2),
        "|", status,
        "| Balance: Rs.", round(current_balance, 2)
    )

successful_deposits = sum(
    1 for t in transaction_history
    if t[0] == "deposit" and t[2] == "Successful deposit"
)

successful_withdrawals = sum(
    1 for t in transaction_history
    if t[0] == "withdraw" and t[2] == "Successful withdrawal"
)

rejected_transactions = sum(
    1 for t in transaction_history
    if t[2].startswith("Rejected")
)

print("\nSuccessful Deposits:", successful_deposits)
print("Successful Withdrawals:", successful_withdrawals)
print("Rejected Transactions:", rejected_transactions)
print("Final Balance: Rs.", round(balance, 2))
print("==================================")
```



```
======== RESTART: C:/Users/YOGITHA/OneDrive/Desktop/5.ATM transaction.py =======
===== WELCOME TO PYTHON ATM =====
Initial Balance: Rs. 5000.0
Enter number of transactions: 1

Transaction 1
Enter deposit or withdraw: deposit
Enter amount: Rs. 1000
Status: Successful deposit
Current Balance: Rs. 6000.0

===== TRANSACTION SUMMARY =====
Deposit | Amount: Rs. 1000.0 | Successful deposit | Balance: Rs. 6000.0

Successful Deposits: 1
Successful Withdrawals: 0
Rejected Transactions: 0
Final Balance: Rs. 6000.0
================================

======== RESTART: C:/Users/YOGITHA/OneDrive/Desktop/5.ATM transaction.py =======
===== WELCOME TO PYTHON ATM =====
Initial Balance: Rs. 5000.0
Enter number of transactions: 2

Transaction 1
Enter deposit or withdraw: withdraw
Enter amount: Rs. 1500
Status: Successful withdrawal
Current Balance: Rs. 3500.0

Transaction 2
Enter deposit or withdraw:
```

10. Conclusion

The ATM Transaction Sequence Simulation program processes deposits and withdrawals, maintains the account balance, enforces withdrawal limits, and rejects invalid transactions. It generates a transaction summary and demonstrates Python variables, operators, conditional statements, loops, lists, tuples, and tuple unpacking.